Samuele Pironaci

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PROFILE

Third-year Computer Science and AI student at the University of Calabria. Most of what I build is backend stuff in Java with Spring Boot, or in Node.js, usually on PostgreSQL or MongoDB, and I write C++ too. Lately I've also been getting into mobile with Kotlin Multiplatform. The part I enjoy most is designing a clean REST API and then building the interface that uses it.

EDUCATION

University of Calabria In progress

Bachelor's degree in Computer Science and Artificial Intelligence

Relevant courses: Data Structures and Algorithms, Discrete Mathematics, Databases, Operating Systems, Web Applications, User Interface Design, Networks and Security, Parallel Algorithms and Distributed Systems, Artificial Intelligence.

PROJECTS

GeoSentinel Sep 2026 - present

Personal project · mobile app and backend

Java 21, Spring Boot 4, PostgreSQL + PostGIS, Redis, MinIO · Kotlin Multiplatform

- A trail safety app for mountain parks. Hikers see the trails on a map, report hazards with a photo, book mountain huts and can send an SOS. Park managers and rescue teams get the reports, open maintenance tickets and send out squads.
- The backend is Spring Boot with a hexagonal architecture (a pragmatic version of it). Trails and hazards live in PostGIS, so the spatial queries run directly in the database.
- Token-based login with refresh tokens that rotate, rate limiting in Redis using Lua scripts, and idempotency keys on SOS and bookings. Errors follow RFC 7807.
- Photos are uploaded straight to MinIO with presigned POST policies.
- The Android and iOS apps share one Kotlin codebase, with MapLibre for the maps and a local cache using ETags. If there's no connection, bookings and SOS wait in an outbox and go out once the phone is back online. Tokens are kept in the Android Keystore and iOS Keychain.
- Integration tests with Testcontainers, 300+ tests on the app, CI on GitHub Actions. In production it runs on Docker Compose with daily backups. I keep the main design decisions written down as ADRs and arc42/C4 docs.

Ricette Facili (recipe blog) Jan 2026

Spring Boot, Angular, PostgreSQL

- Recipe blog with a Spring Boot REST backend, a separate Angular frontend and PostgreSQL for the data.
- I designed the REST API and built a responsive interface on top of it.

Food Delivery+ Sep 2025

Java, JDBC, SQL

- Java desktop app built with MVC plus the DAO and Singleton patterns, so the business logic stays out of the UI and the data layer.
- Data access through JDBC and plain SQL. I paid attention to OOP principles to keep the code easy to maintain.

Multi-agent AI: Tron edition

Answer Set Programming, DLV2, EmbASP, MVC

- Real-time game where four AI agents compete on a grid. The game state sits in an MVC model, and each agent decides its moves in a separate logic layer.
- The agents' logic is written in ASP and solved by DLV2 through EmbASP. At every tick the solver goes through thousands of possible moves to avoid borders, trails and the other agents, and integrity constraints rule out U-turns and head-on crashes.

TECHNICAL SKILLS

Languages (advanced)	Java, C/C++, Python, SQL
Languages (basic)	Kotlin, TypeScript, Rust (learning)
Backend	Spring Boot (Web, Security, Data JPA), Node.js, Express, REST
Mobile & frontend	Kotlin Multiplatform, React, Angular
Databases & storage	PostgreSQL, PostGIS, MySQL, MongoDB, Redis, MinIO (S3)
DevOps	Docker, Docker Compose, GitHub Actions (CI/CD)
Testing	JUnit, Testcontainers
Parallel computing	MPI, POSIX Threads, CUDA (learning)
Tools	Git, GitHub, IntelliJ IDEA

WORK EXPERIENCE

Waiter and cashier, Chocolat, Soverato (part-time) Jun - Aug 2023

Waiter, Drop, Soverato (part-time) Jun - Aug 2022

SOFT SKILLS

I like solving problems in a practical way and I pick up new tools quickly. I'm open to feedback and code review, and I try to communicate clearly when working in a team.